

Online Resources

A curated collection of online platforms, courses, and resources for learning data structures and algorithms.

Interactive Learning Platforms

LeetCode

URL: leetcode.com (<https://leetcode.com>)

Type: Problem solving platform

Cost: Free (Premium: \$35/month)

Features:

- 2500+ coding problems
- Difficulty levels: Easy, Medium, Hard
- Company-specific question lists
- Discussion forums with solutions
- Mock interviews
- Contest competitions

Best for: Interview preparation, practical problem-solving

Tips:

- Start with Easy problems
 - Focus on patterns, not memorization
 - Read discussion after solving
 - Track progress with tags
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HackerRank

URL: hackerrank.com (<https://hackerrank.com>)

Type: Learning and practice platform

Cost: Free

Features:

- Structured learning tracks
- Skill certifications
- Interview preparation kits
- Company challenges
- 10 Days of Statistics/JavaScript/etc.

Best for: Beginners learning fundamentals, building resume

Tracks to follow:

- Data Structures
- Algorithms
- Python
- Interview Preparation Kit

CodeSignal

URL: codesignal.com (https://codesignal.com)

Type: Practice and assessment platform

Cost: Free

Features:

- Arcade mode for fun challenges
- Interview practice
- Company-specific assessments
- Skill-based rankings
- Detailed feedback

Best for: Gamified learning, assessment practice

AlgoExpert

URL: algoexpert.io (https://algoexpert.io)

Type: Curated interview prep

Cost: \$99-\$164 (one-time)

Features:

- 160+ curated problems
- Video explanations for each
- Space/time complexity analysis
- Multiple language solutions
- SystemsExpert included

Best for: Focused interview prep with guided explanations



Video Courses & Channels

YouTube Channels

1. Abdul Bari

Channel: [Abdul Bari](https://www.youtube.com/channel/UCZCFT11CWBi3MHNIGf019nw) (https://www.youtube.com/channel/UCZCFT11CWBi3MHNIGf019nw)

Focus: Algorithms with clear visual explanations

Best for: Understanding algorithm internals

2. MIT OpenCourseWare

Course: [Introduction to Algorithms](https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/) (https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/)

Focus: Academic, comprehensive coverage

Best for: Theoretical depth

3. mycodeschool

Channel: [mycodeschool](https://www.youtube.com/user/mycodeschool) (https://www.youtube.com/user/mycodeschool)

Focus: Data structures and algorithms fundamentals

Best for: Clear, concise explanations of basics

4. William Fiset

Channel: [William Fiset](https://www.youtube.com/c/WilliamFiset-videos) (https://www.youtube.com/c/WilliamFiset-videos)

Focus: Graph theory, data structures

Best for: Deep dives into specific topics

5. Back To Back SWE

Channel: [Back To Back SWE](https://www.youtube.com/c/BackToBackSWE) (https://www.youtube.com/c/BackToBackSWE)

Focus: Interview problem walkthroughs

Best for: Learning problem-solving approaches

6. NeetCode

Channel: [NeetCode](https://www.youtube.com/c/NeetCode) (https://www.youtube.com/c/NeetCode)

Focus: LeetCode problem explanations

Best for: Visual LeetCode solutions

Online Courses

Coursera

1. Algorithms Specialization by Stanford

- **Instructor:** Tim Roughgarden
- **Duration:** ~4 months
- **Cost:** \$49/month (7-day free trial)
- **Covers:** Divide and conquer, graphs, greedy, dynamic programming

2. Data Structures and Algorithms by UC San Diego

- **Duration:** ~6 months
- **Cost:** \$49/month
- **Covers:** Comprehensive DSA with programming assignments

Udemy

1. Master the Coding Interview: Data Structures + Algorithms

- **Instructor:** Andrei Neagoie
- **Duration:** 19 hours
- **Cost:** ~\$15-20 (frequent sales)
- **Covers:** Big O, DSA, interview techniques

2. Python for Data Structures, Algorithms, and Interviews!

- **Instructor:** Jose Portilla
- **Duration:** 16.5 hours
- **Cost:** ~\$15-20
- **Covers:** Python-specific DSA

edX

1. Algorithms and Data Structures by Microsoft

- **Cost:** Free (verified certificate: \$99)
- **Covers:** Fundamental algorithms and data structures



Documentation & References

Official Documentation

1. Python Documentation

- **URL:** [docs.python.org](https://docs.python.org/3/) (https://docs.python.org/3/)
- **Focus:** Built-in data structures
- **Key sections:** Data Structures, collections module

2. Python Time Complexity

- **URL:** wiki.python.org/moin/TimeComplexity (https://wiki.python.org/moin/TimeComplexity)
- **Focus:** Big O for Python operations

3. NumPy Documentation

- **URL:** numpy.org/doc (https://numpy.org/doc/)
- **Focus:** Array operations and algorithms

4. Pandas Documentation

- **URL:** pandas.pydata.org/docs (https://pandas.pydata.org/docs/)
- **Focus:** Data structure manipulation

Algorithm Visualizations

1. VisuAlgo

- **URL:** visualgo.net (https://visualgo.net)
- **Features:** Animated visualizations of algorithms
- **Topics:** Sorting, trees, graphs, dynamic programming
- **Best for:** Visual learners

2. Algorithm Visualizer

- **URL:** algorithm-visualizer.org (https://algorithm-visualizer.org)
- **Features:** Interactive algorithm animations
- **Best for:** Understanding algorithm flow

3. USF Data Structure Visualizations

- **URL:** [usfca.edu/cs/galles/visualization](https://www.cs.usfca.edu/~galles/visualization/) (https://www.cs.usfca.edu/~galles/visualization/)
- **Features:** Step-by-step animations
- **Best for:** Learning data structure operations

4. Toptal Sorting Algorithms Animations

- **URL:** [toptal.com/developers/sorting-algorithms](https://www.toptal.com/developers/sorting-algorithms) (https://www.toptal.com/developers/sorting-algorithms)
- **Features:** Compare sorting algorithms visually
- **Best for:** Understanding sorting differences

Websites & Blogs

Educational Sites

1. GeeksforGeeks

- **URL:** [geeksforgeeks.org](https://www.geeksforgeeks.org) (<https://www.geeksforgeeks.org>)
- **Features:** Tutorials, problems, company interview questions
- **Best for:** Quick reference and examples

2. Tutorialspoint

- **URL:** [tutorialspoint.com/data_structures_algorithms](https://www.tutorialspoint.com/data_structures_algorithms/) (https://www.tutorialspoint.com/data_structures_algorithms/)
- **Features:** Structured tutorials
- **Best for:** Beginners learning fundamentals

3. Programiz

- **URL:** [programiz.com/dsa](https://www.programiz.com/dsa) (<https://www.programiz.com/dsa>)
 - **Features:** Beginner-friendly explanations
 - **Best for:** Clear, simple tutorials
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Technical Blogs

1. Baeldung (Computer Science)

- **URL:** [baeldung.com/cs](https://www.baeldung.com/cs/) (<https://www.baeldung.com/cs/>)
- **Focus:** In-depth algorithm explanations

2. Real Python

- **URL:** realpython.com (<https://realpython.com>)
- **Focus:** Python-specific tutorials

3. Towards Data Science (Medium)

- **URL:** towardsdatascience.com (<https://towardsdatascience.com>)
 - **Focus:** Algorithms and their applications
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Competitive Programming

Platforms

1. Codeforces

- **URL:** codeforces.com (<https://codeforces.com>)
- **Features:** Regular contests, rating system
- **Best for:** Advanced competitive programming

2. AtCoder

- **URL:** atcoder.jp (<https://atcoder.jp>)
- **Features:** Weekly contests, educational content
- **Best for:** Learning through competition

3. TopCoder

- **URL:** [topcoder.com](https://www.topcoder.com) (<https://www.topcoder.com>)
- **Features:** Challenges, tutorials, competitions
- **Best for:** Algorithm competitions

4. Project Euler

- **URL:** projecteuler.net (<https://projecteuler.net>)
 - **Features:** Mathematical/computational problems
 - **Best for:** Math + programming challenges
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Mobile Apps

1. Enki (iOS/Android)

- **Focus:** Daily coding challenges
- **Cost:** Free with premium option

2. SoloLearn (iOS/Android)

- **Focus:** Interactive coding lessons
- **Cost:** Free

3. Mimo (iOS/Android)

- **Focus:** Gamified coding education
 - **Cost:** Free with premium
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Gamified Learning

1. CodinGame

- **URL:** [codingame.com](https://www.codingame.com) (<https://www.codingame.com>)
- **Features:** Learn by playing games
- **Best for:** Making learning fun

2. CheckiO

- **URL:** checkio.org (<https://checkio.org>)
- **Features:** Python/TypeScript game-based learning
- **Best for:** Interactive problem-solving

3. Codewars

- **URL:** [codewars.com](https://www.codewars.com) (<https://www.codewars.com>)
 - **Features:** Kata challenges with ranking
 - **Best for:** Practice with community solutions
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Interview Preparation

1. Blind 75

- **Resource:** Collection of 75 essential LeetCode problems
- **Best for:** Focused interview prep
- **Access:** Search "Blind 75 LeetCode"

2. NeetCode 150

- **URL:** neetcode.io (https://neetcode.io)
- **Features:** Curated list with video solutions
- **Best for:** Structured interview preparation

3. Tech Interview Handbook

- **URL:** [techinterviewhandbook.org](https://www.techinterviewhandbook.org) (https://www.techinterviewhandbook.org)
 - **Features:** Comprehensive interview guide
 - **Best for:** Complete interview preparation
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Practice Tools

1. Repl.it

- **URL:** replit.com (https://replit.com)
- **Features:** Online IDE, collaborative coding
- **Best for:** Quick practice without setup

2. Python Tutor

- **URL:** pythontutor.com (https://pythontutor.com)
- **Features:** Visualize code execution step-by-step
- **Best for:** Understanding code flow

3. LeetCode Playground

- **Features:** Quick algorithm testing
 - **Best for:** Testing solutions
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GitHub Repositories

1. TheAlgorithms/Python

- **URL:** github.com/TheAlgorithms/Python
- **Features:** Implementations of all algorithms

2. donnemartin/system-design-primer

- **URL:** github.com/donnemartin/system-design-primer
- **Features:** System design interview prep

3. jwasham/coding-interview-university

- **URL:** github.com/jwasham/coding-interview-university
 - **Features:** Complete study plan for becoming a software engineer
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Tips for Using Online Resources

1. **Don't spread too thin:** Pick 2-3 resources and stick with them
2. **Practice consistently:** Daily practice beats marathon sessions
3. **Active learning:** Code along, don't just watch/read
4. **Track progress:** Use spreadsheets or apps to monitor

5. **Join communities:** Discord, Reddit for motivation and help

6. **Revisit concepts:** Spaced repetition improves retention



Suggested Learning Path

Week 1-2: Fundamentals

- HackerRank Data Structures track
- Abdul Bari videos for theory

Week 3-6: Practice

- LeetCode Easy problems
- VisuAlgo for visualizations

Week 7-10: Intermediate

- LeetCode Medium problems
- Coursera Algorithms course

Week 11+: Advanced & Interview Prep

- NeetCode 150
- Mock interviews
- System design



Related Resources

- [Book Recommendations](#) (book_recommendations.md)
 - [Interview Prep Guide](#) (interview_prep.md)
 - [Study Plan](#) (study_plan.md)
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